

8.3 Isomorphism

In this section we establish a fundamental connection between real finite-dimensional vector spaces and the Euclidean space R^n . This connection is theoretically important and has practical value: it allows vector computations in general n -dimensional spaces to be performed by working with vectors in R^n .

Isomorphism

Although many theorems in this text concern R^n exclusively, every real n -dimensional vector space has the same algebraic structure as R^n , even if its vectors are not n -tuples. To make this precise, we need the following definition.

Definition 1

A linear transformation $T : V \rightarrow W$ that is both one-to-one and onto is called an **isomorphism**, and W is said to be **isomorphic** to V .

The following diagram illustrates an isomorphism between P_2 and R^3 via $T(c_0 + c_1x + c_2x^2) = (c_0, c_1, c_2)$; operations on one side correspond exactly to operations on the other:

$$\begin{array}{ccc} k(c_0 + c_1x + c_2x^2) & \xrightarrow{T} & k(c_0, c_1, c_2) \\ \updownarrow & & \updownarrow \\ kc_0 + kc_1x + kc_2x^2 & \xrightarrow{T} & (kc_0, kc_1, kc_2) \end{array}$$

$$(c_0 + d_0) + (c_1 + d_1)x + (c_2 + d_2)x^2 \xrightarrow{T} (c_0 + d_0, c_1 + d_1, c_2 + d_2)$$

The algebraic structure of P_2 is mirrored exactly in R^3 , and vice versa via T^{-1} .

Theorem 8.3.1

Every real n -dimensional vector space is isomorphic to R^n .

Proof. Let V be a real n -dimensional vector space with ordered basis $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, and define the coordinate map $T : V \rightarrow R^n$ by

$$T(\mathbf{u}) = (\mathbf{u})_S = (k_1, k_2, \dots, k_n) \tag{2}$$

where $\mathbf{u} = k_1\mathbf{v}_1 + \dots + k_n\mathbf{v}_n$. Writing $\mathbf{v} = d_1\mathbf{v}_1 + \dots + d_n\mathbf{v}_n$:

$$T(c\mathbf{u}) = c(k_1, \dots, k_n) = cT(\mathbf{u}) \quad T(\mathbf{u} + \mathbf{v}) = (k_1 + d_1, \dots, k_n + d_n) = T(\mathbf{u}) + T(\mathbf{v})$$

so T is linear. Since distinct vectors have distinct coordinate tuples, $T(\mathbf{u}) \neq T(\mathbf{v})$ whenever $\mathbf{u} \neq \mathbf{v}$, so T is one-to-one. For any $(k_1, \dots, k_n) \in R^n$ the vector $k_1 \mathbf{v}_1 + \dots + k_n \mathbf{v}_n$ maps to it, so T is onto. ■

Theorem 8.3.2

If S is an ordered basis for a vector space V , then the coordinate map

$$\mathbf{u} \xrightarrow{T} (\mathbf{u})_S$$

is an isomorphism between V and R^n .

Remark. Coordinate maps depend on the order of basis vectors; Theorem 8.3.2 describes $n!$ different isomorphisms (one per ordering of the n basis vectors).

EXAMPLE 1 | Natural Isomorphism Between P_{n-1} and R^n

By Theorem 8.3.2, the coordinate map with respect to the standard basis $\{1, x, \dots, x^{n-1}\}$ defines the **natural isomorphism** between P_{n-1} and R^n :

$$a_0 + a_1x + \dots + a_{n-1}x^{n-1} \xrightarrow{T} (a_0, a_1, \dots, a_{n-1})$$

EXAMPLE 2 | Natural Isomorphism Between M_{22} and R^4

The coordinate map for M_{22} with respect to the standard basis gives the **natural isomorphism** between M_{22} and R^4 :

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{T} (a, b, c, d)$$

More generally, the analogous map is a natural isomorphism between M_{mn} and R^{mn} .

EXAMPLE 3 | Differentiation by Matrix Multiplication

(Calculus Required)

The differentiation $D : P_3 \rightarrow P_2$ maps $a_0 + a_1x + a_2x^2 + a_3x^3 \xrightarrow{D} a_1 + 2a_2x + 3a_3x^2$. Under the natural isomorphisms $P_3 \cong R^4$ and $P_2 \cong R^3$, this corresponds to

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \end{bmatrix}$$

For example, $\frac{d}{dx}(2 + x + 4x^2 - x^3) = 1 + 8x - 3x^2$ is reproduced by

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -3 \end{bmatrix}$$

This approach is useful for numerical algorithms that compute derivatives.

EXAMPLE 4 | Working with Isomorphisms

Use the natural isomorphism between P_5 and R^6 to determine whether the following polynomials are linearly independent:

$$\mathbf{p}_1 = 1 + 2x - 3x^2 + 4x^3 + x^5$$

$$\mathbf{p}_2 = 1 + 3x - 4x^2 + 6x^3 + 5x^4 + 4x^5$$

$$\mathbf{p}_3 = 3 + 8x - 11x^2 + 16x^3 + 10x^4 + 9x^5$$

SOLUTION: Form the matrix A whose rows are the coordinate vectors under the natural isomorphism:

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 1 & 3 & -4 & 6 & 5 & 4 \\ 3 & 8 & -11 & 16 & 10 & 9 \end{bmatrix}$$

Row-reducing gives

$$R = \begin{bmatrix} 1 & 2 & -3 & 4 & 0 & 1 \\ 0 & 1 & -1 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

R has only two nonzero rows, so the row space of A is two-dimensional, meaning the rows of A are linearly dependent. Therefore $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ are linearly dependent.

Inner Product Space Isomorphisms

When V is a real n -dimensional inner product space, both V and R^n carry geometric as well as algebraic structure. An isomorphism $T : V \rightarrow W$ between inner product spaces is called an **inner product space isomorphism** if

$$\langle T(\mathbf{u}), T(\mathbf{v}) \rangle = \langle \mathbf{u}, \mathbf{v} \rangle \quad \text{for all } \mathbf{u}, \mathbf{v} \in V$$

(inner product on the left is in W ; on the right, in V).

To preserve geometric structure an isomorphism must preserve inner products, since length, angle, and orthogonality all depend on the inner product. The following analog of Theorem 8.3.2 provides an important method for obtaining inner product space isomorphisms.

Theorem 8.3.3

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an ordered orthonormal basis for a real n -dimensional inner product space V , then the coordinate map $\mathbf{u} \xrightarrow{T} (\mathbf{u})_S$ is an inner product space isomorphism between V and R^n with the Euclidean inner product.

EXAMPLE 5 | An Inner Product Space Isomorphism

The natural isomorphism $T : P_{n-1} \rightarrow R^n$ of Example 1 maps $a_0 + a_1x + \dots + a_{n-1}x^{n-1} \mapsto (a_0, a_1, \dots, a_{n-1})$. The standard basis $\{1, x, \dots, x^{n-1}\}$ is orthonormal under the standard inner product on P_{n-1} , so by Theorem 8.3.3, T is an inner product space isomorphism. To verify directly: for $\mathbf{p} = a_0 + \dots + a_{n-1}x^{n-1}$

and $\mathbf{q} = b_0 + \cdots + b_{n-1}x^{n-1}$,

$$\langle \mathbf{p}, \mathbf{q} \rangle = a_0b_0 + a_1b_1 + \cdots + a_{n-1}b_{n-1}$$

which equals the Euclidean inner product of $(a_0, \dots, a_{n-1})$ and $(b_0, \dots, b_{n-1})$ in R^n . ✓

EXAMPLE 6 | A Notational Matter

Let R^n denote n -tuples in comma-delimited form and M_n denote real $n \times 1$ column matrices, with inner products $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v}$ and $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v}$ respectively. The map

$$T : R^n \rightarrow M_n, \quad (v_1, v_2, \dots, v_n) \mapsto \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

is an inner product space isomorphism, confirming that R^n and M_n are essentially the same inner product space — a fact used throughout this text.

Exercise Set 8.3

In Exercises 1–8, state whether the transformation is an isomorphism. No proof required.

1. $c_0 + c_1x \mapsto (c_0 - c_1, c_1)$ from P_1 to R^2 .
3. $a + bx + cx^2 + dx^3 \mapsto \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ from P_3 to M_{22} .
5. $(a, b, c, d) \mapsto a + bx + cx^2 + (d + 1)x^3$ from R^4 to P_3 .
6. $A \mapsto A^T$ from M_{nn} to M_{nn} .
7. $c_1 \sin x + c_2 \cos x \mapsto (c_1, c_2)$ from the subspace of $C(-\infty, \infty)$ spanned by $\{\sin x, \cos x\}$ to R^2 .
9. **a.** Find an isomorphism between the vector space of all 3×3 symmetric matrices and R^6 .
b. Find two different isomorphisms between the vector space of all 2×2 matrices and R^4 .
10. **a.** Find an isomorphism between the vector space of all polynomials in P_3 satisfying $p(0) = 0$ and R^3 .

In Exercises 11–12, use Theorem 8.2.2 to determine whether $T_A : R^3 \rightarrow R^3$ is an isomorphism.

$$11. A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}$$

In Exercises 13–14, find the dimension n of the solution space W of $A\mathbf{x} = \mathbf{0}$, then construct an isomorphism between W and R^n .

13. $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \end{bmatrix}$

In Exercises 15–16, determine whether the transformation is an isomorphism from M_{22} to R^4 .

15. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{pmatrix} a \\ a+b \\ a+b+c \\ a+b+c+d \end{pmatrix}$

17. Is R^2 isomorphic to the xy -plane in R^3 ? Justify your answer.

19. Let $T : P_2 \rightarrow M_{22}$ be defined by

$$T(p(x)) = \begin{bmatrix} p(0) & p(1) \\ p(1) & p(0) \end{bmatrix}$$

Is T an isomorphism? Justify your answer.

Working with Proofs

22. Prove: if $T : V \rightarrow W$ is an isomorphism, then $T^{-1} : W \rightarrow V$ is also an isomorphism.

25. Prove that an inner product space isomorphism preserves angles and distances: the angle between $T(\mathbf{u})$ and $T(\mathbf{v})$ in W equals the angle between $\mathbf{u}$ and $\mathbf{v}$ in V , and $\|T(\mathbf{u}) - T(\mathbf{v})\|_W = \|\mathbf{u} - \mathbf{v}\|_V$.